MIREIA RIBERA | ACCESSIBILITAT DIGITAL | EXPERIÈNCIA D'USUARI | VISUALITZACIÓ DE DADES

Presentation and Visualization. Interaction



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Contents

1. Why interaction?
2. Data manipulation or encoding
3. Exploration and navigation
4. Problem solving
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"Overview first,

zoom and filter,

details on demand"

Schneiderman's Mantra

Objectives

1. Know the main interaction techniques and their properties.
2. Systematize the different objectives, triggers and actions of the different types of interaction.
3. Have criteria for including one type or another of interaction in a visualization.

Types of interaction

Manipulation



Exploration



Problem solving



What goal do we pursue with the interaction?

1. Change *how* data or *what* data is displayed
2. Moving around in the visual space of data
3. Answer questions such as why? or what if...?

PROS y CONS

- ✓ Interaction is the greatest contribution of the digital world to visualization.
- ✓ Interaction means a cognitive effort for the user.

Manipulation of displayed data or encoding



What to show



Filter, add or select

Select the variables or attributes to display.



Layers

In maps, display just part of the total information. Avoid overloading.

How to show it



Order

Display information in ascending or descending order for better comparison.



Change position

The user repositions the elements.



Change style

The user customizes the way the information is displayed.



Project

Change the transformation model from 3D to 2D.

What to show: Filter, add or select

Objective

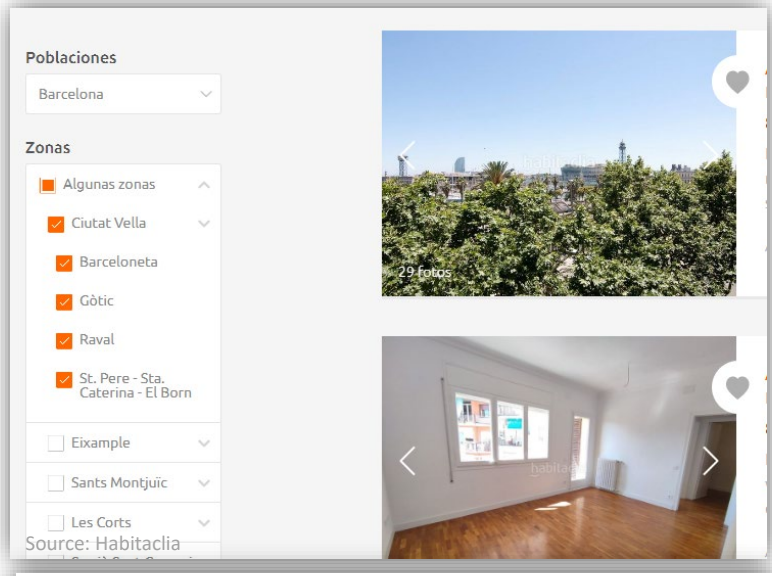
Select the variables or attributes to display.

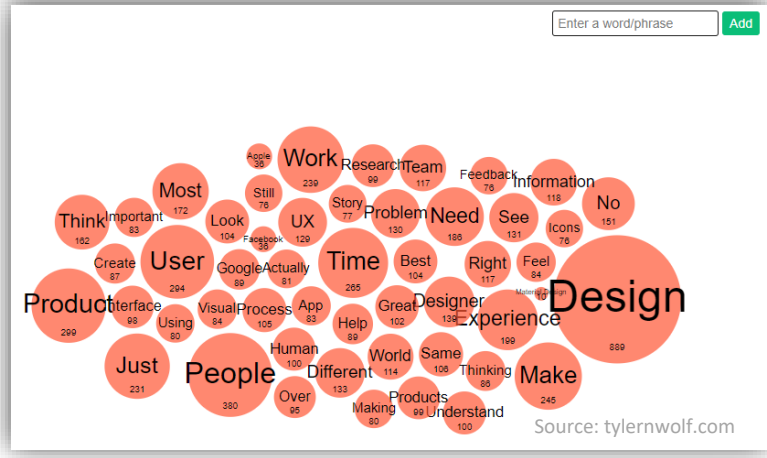
Trigger

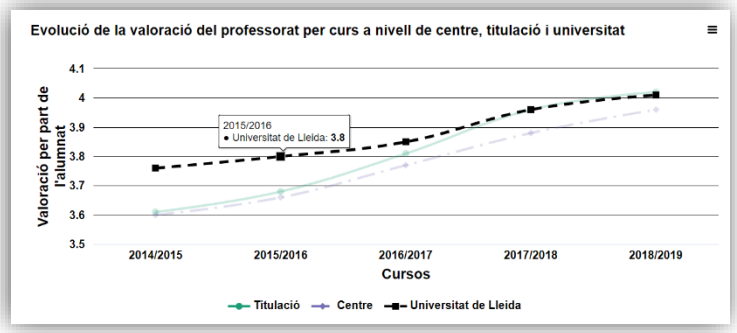
Select an option, select an area, type text.

Action

Selected items are highlighted, unselected items disappear or fade.







What to show: Layers

Objective

In maps, display a type of information. Avoid overloading.

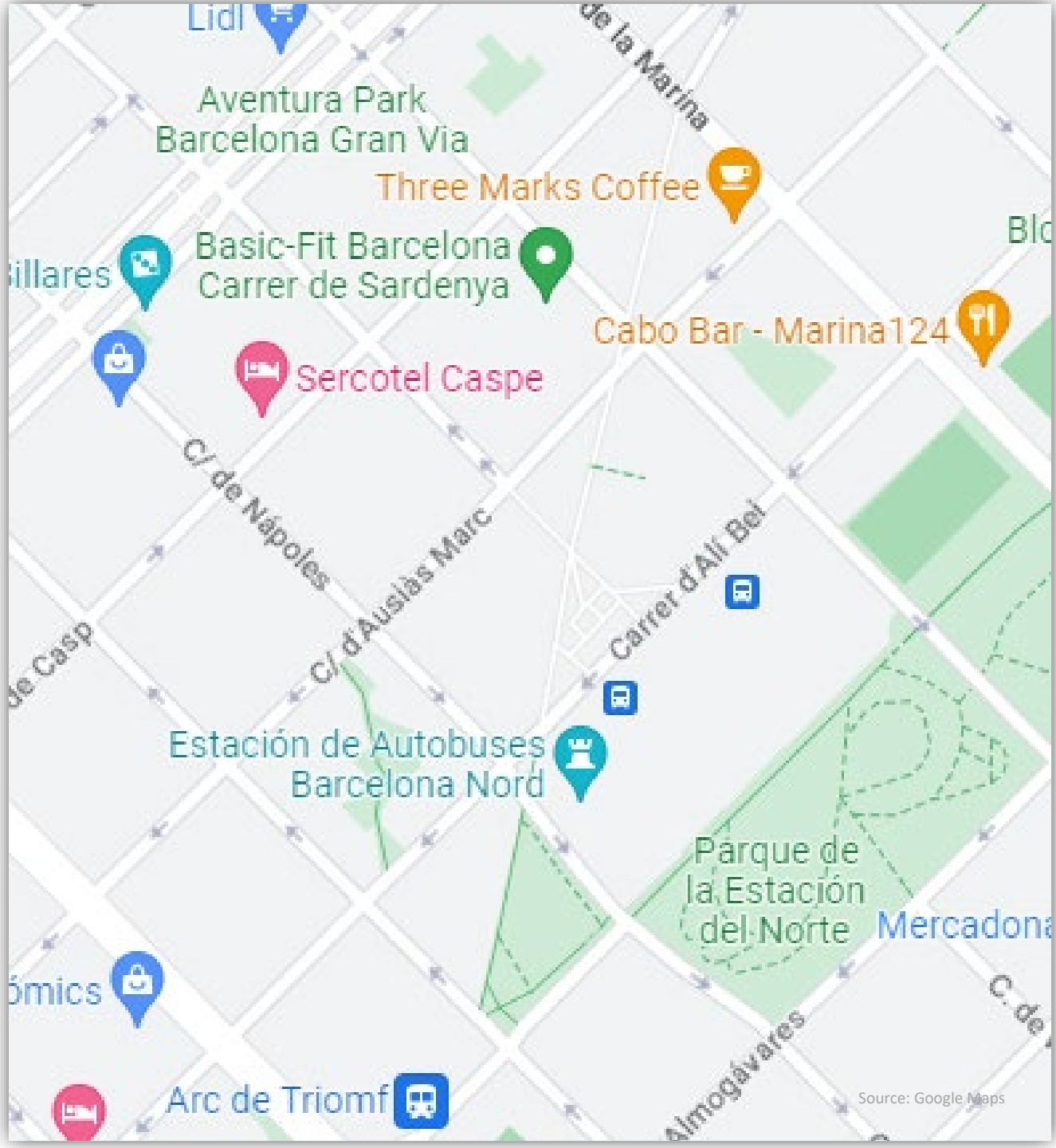
Trigger

Select an option.

Action

A new layer of information is displayed in the visualization.

Related to: Filter and add.



How to show: Order

Objective

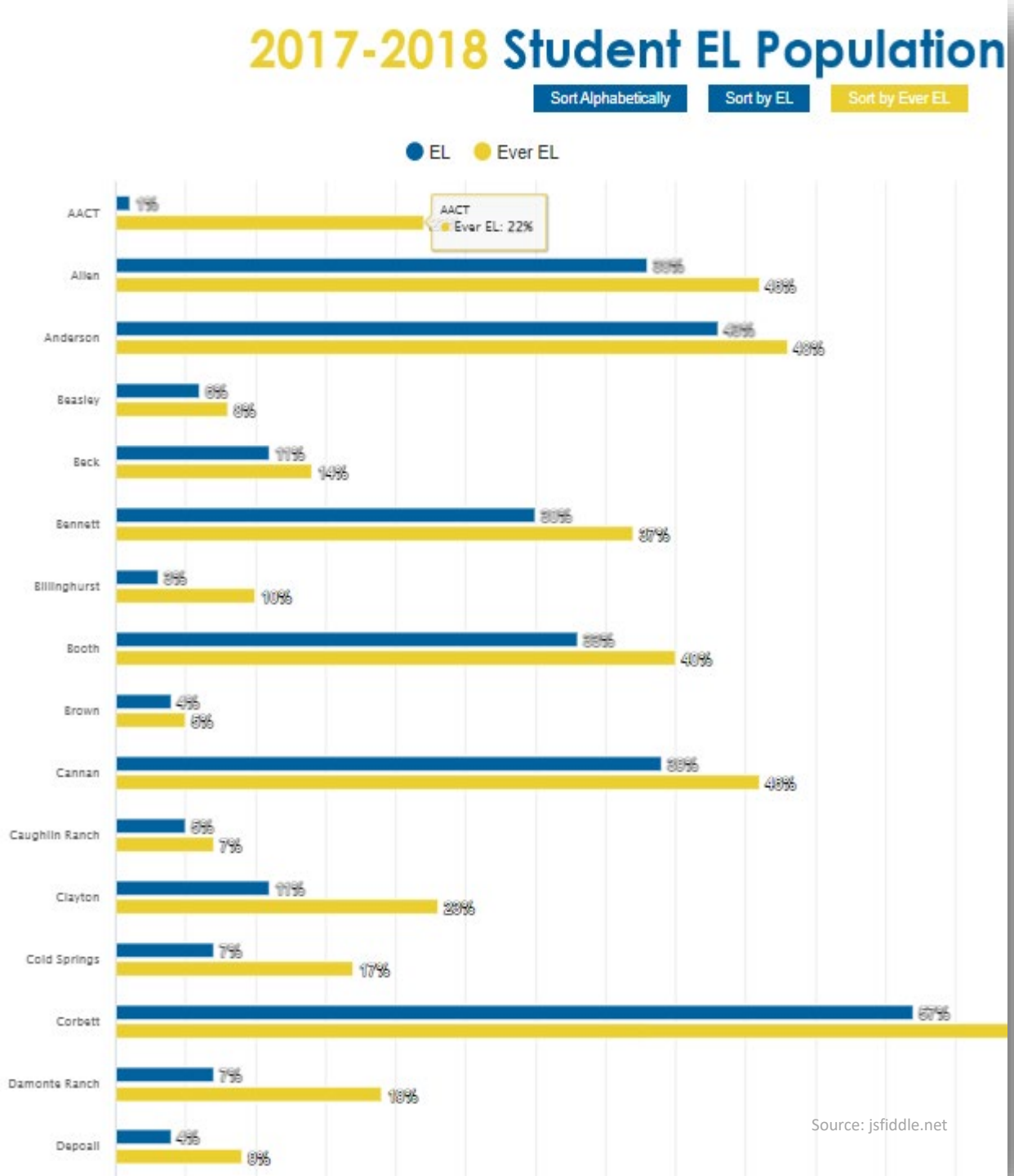
Display information in ascending or descending order to better compare it.

Trigger

Sort button or specific "widget".

Action

Items are reordered.



How to display: Change style

Objective

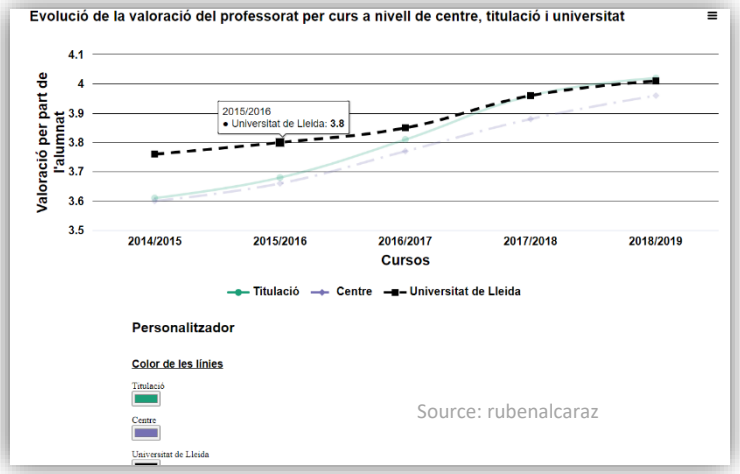
Change the data encoding, colors, line type...
even the chart type.
Sometimes it's the solution for accessibility.

Trigger

Select an option.

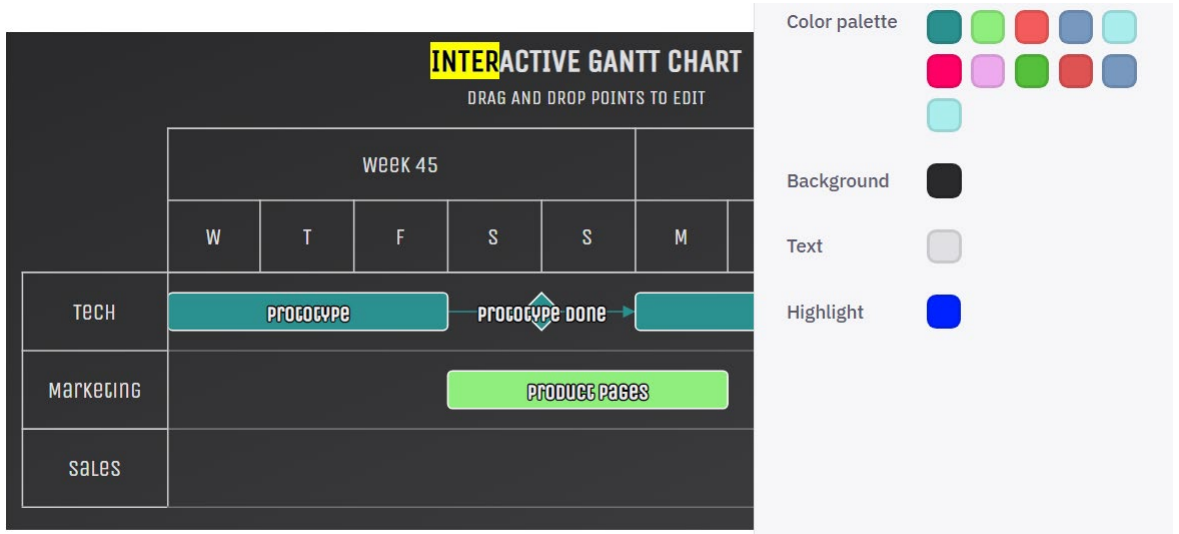
Action

The display is renewed with the new encoding.



Source: rubenalcaraz

Source: Highcharts



How to show: Change position

Objective

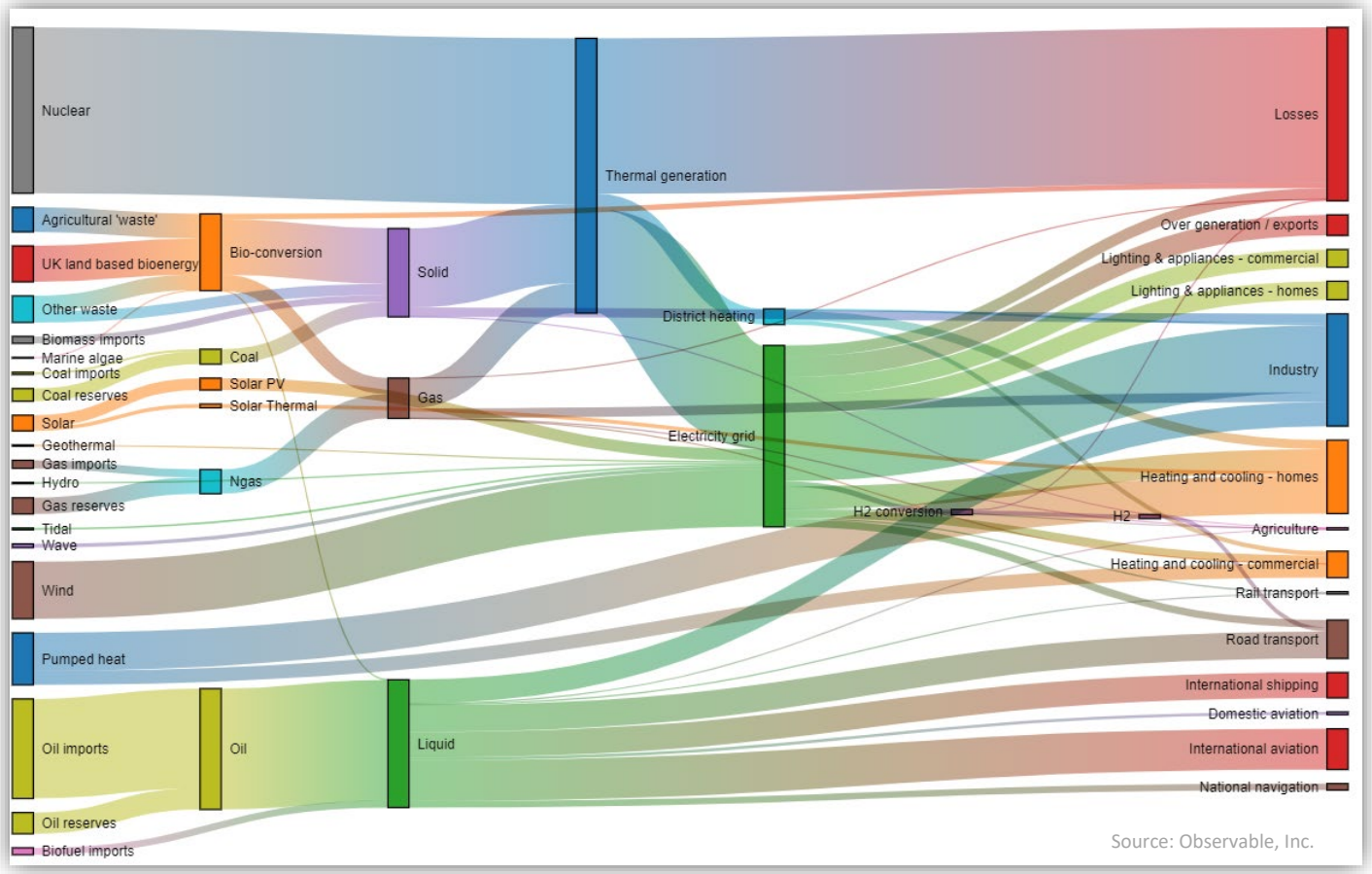
The user repositions the elements.

Trigger

Click and drag.

Action

The selected items change their position. Other elements can be relocated as well.



How to show: Project

Objective

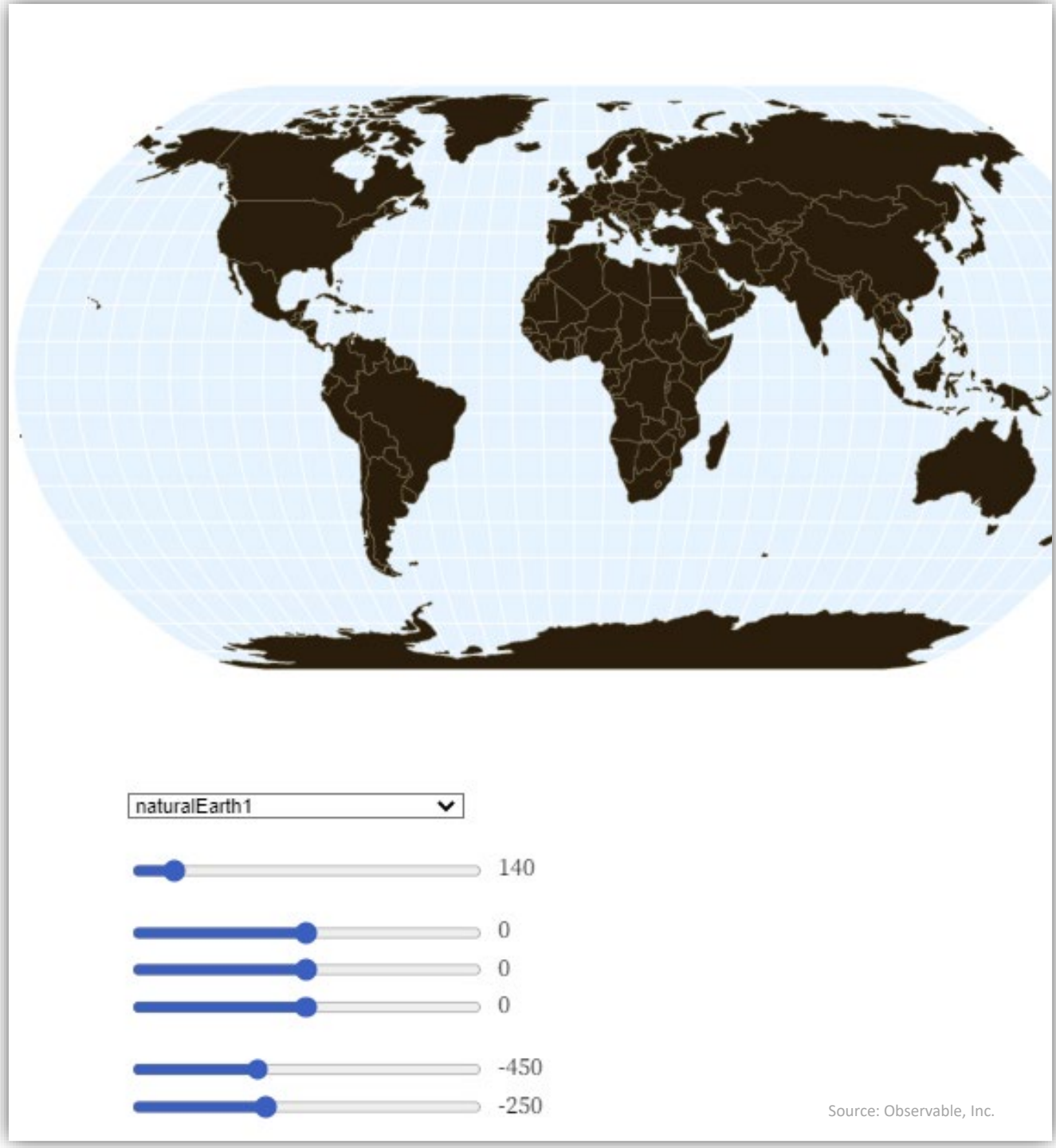
Change the transformation model from 3D to 2D.

Trigger

Specific buttons.

Action

The projection of the displayed items changes to the new projection chosen.



Exploration and navigation



Zoom



Optical zoom

Increase or decrease the displayed area.



Semantic zoom

Similar to optical zoom. When it is increased, new details appear.



Drill down

We move forward/backward in a hierarchy to see more detailed information.

Navigation



Change the point of view

Use a new perspective. May derive in cutting or slicing the object.



Drag (pan)

Navigate the display area.



Time as a third dimension

We change the visualization over time.

Zoom: Optical zoom

Objective

Increase or decrease the area shown.

Trigger

Action with mouse or keyboard or zoom or pinch gesture.

Action

The view is scaled.

Related to:

Semantic zoom.



Zoom: Semantic zoom

Objective

Similar to optical zoom. When it is increased, new details appear.

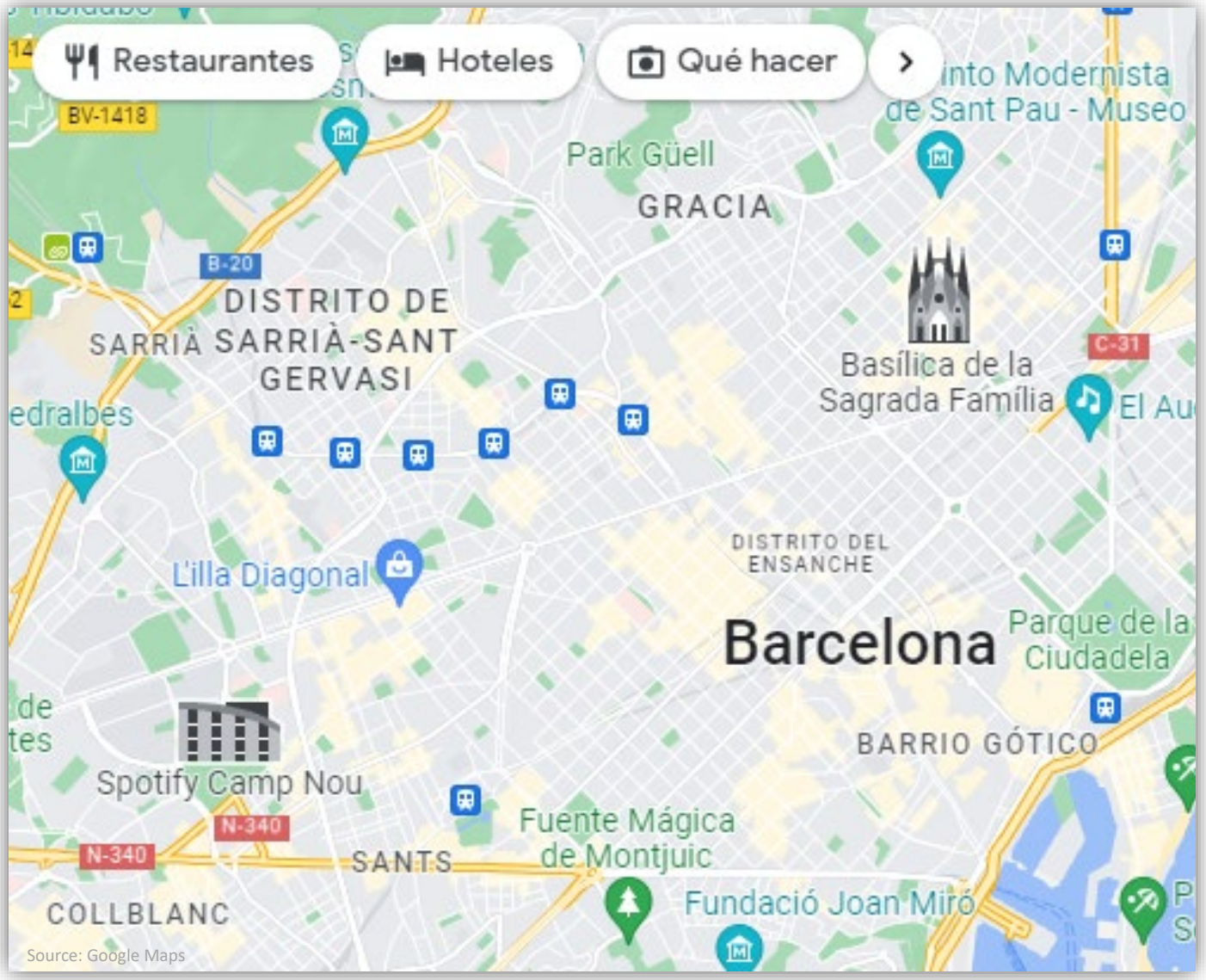
Trigger

Action with mouse or keyboard or zoom or pinch gesture.

Action

The view is scaled. New details appear.

Related to: Optical zoom.



Zoom: Drill down

Objective

We move forward in a hierarchy to see more detailed information.

Trigger

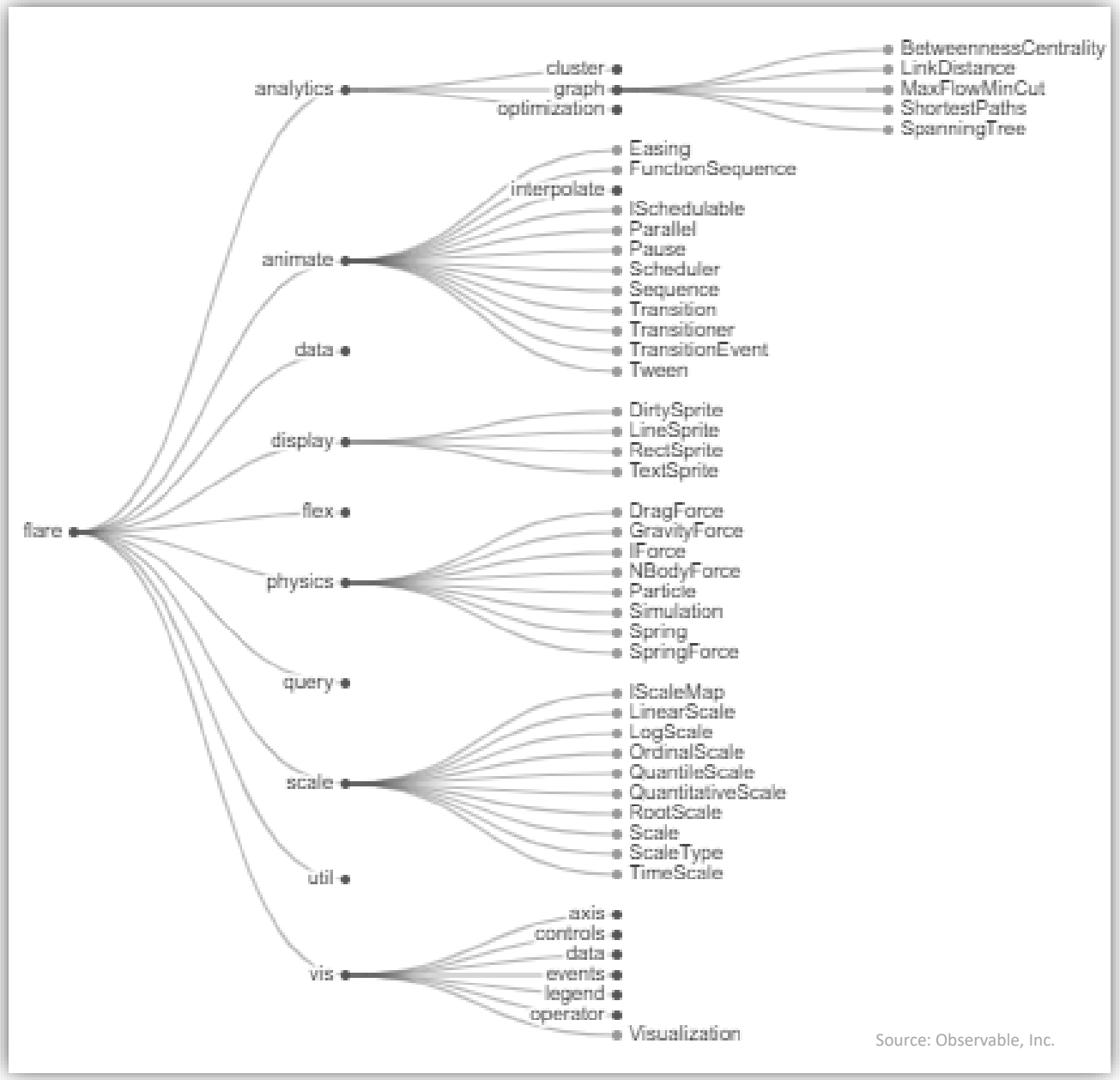
Select one of the categories.

Action

The selected category expands to show its descendants.

Related to:

Problem solving.



Source: Observable, Inc.

Navigation: Change point of view



Objective

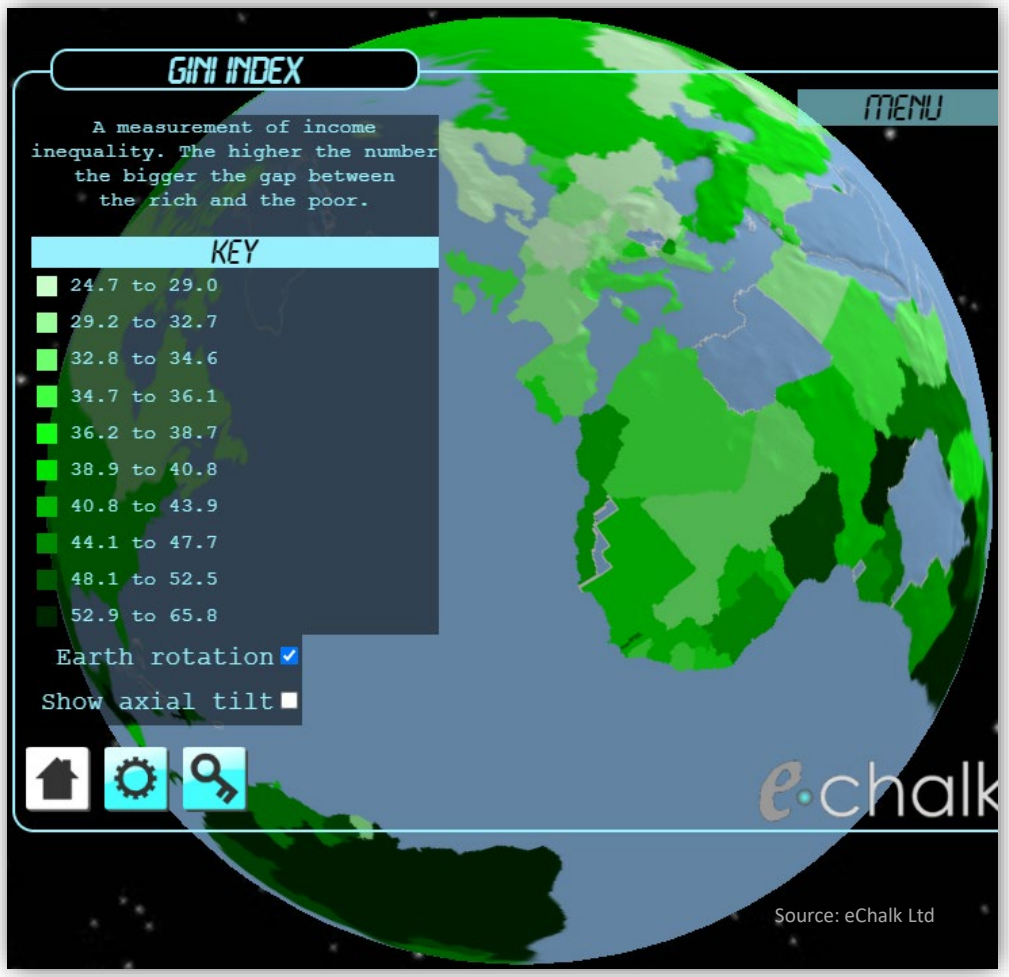
Use a new perspective. It can mean cutting or slicing the object.

Trigger

Direct manipulation with the mouse, with the arrows or dragging gesture.

Action

The orientation of the displayed objects varies.



Navigation: Drag(pan)

Objective

Navigating the display area.

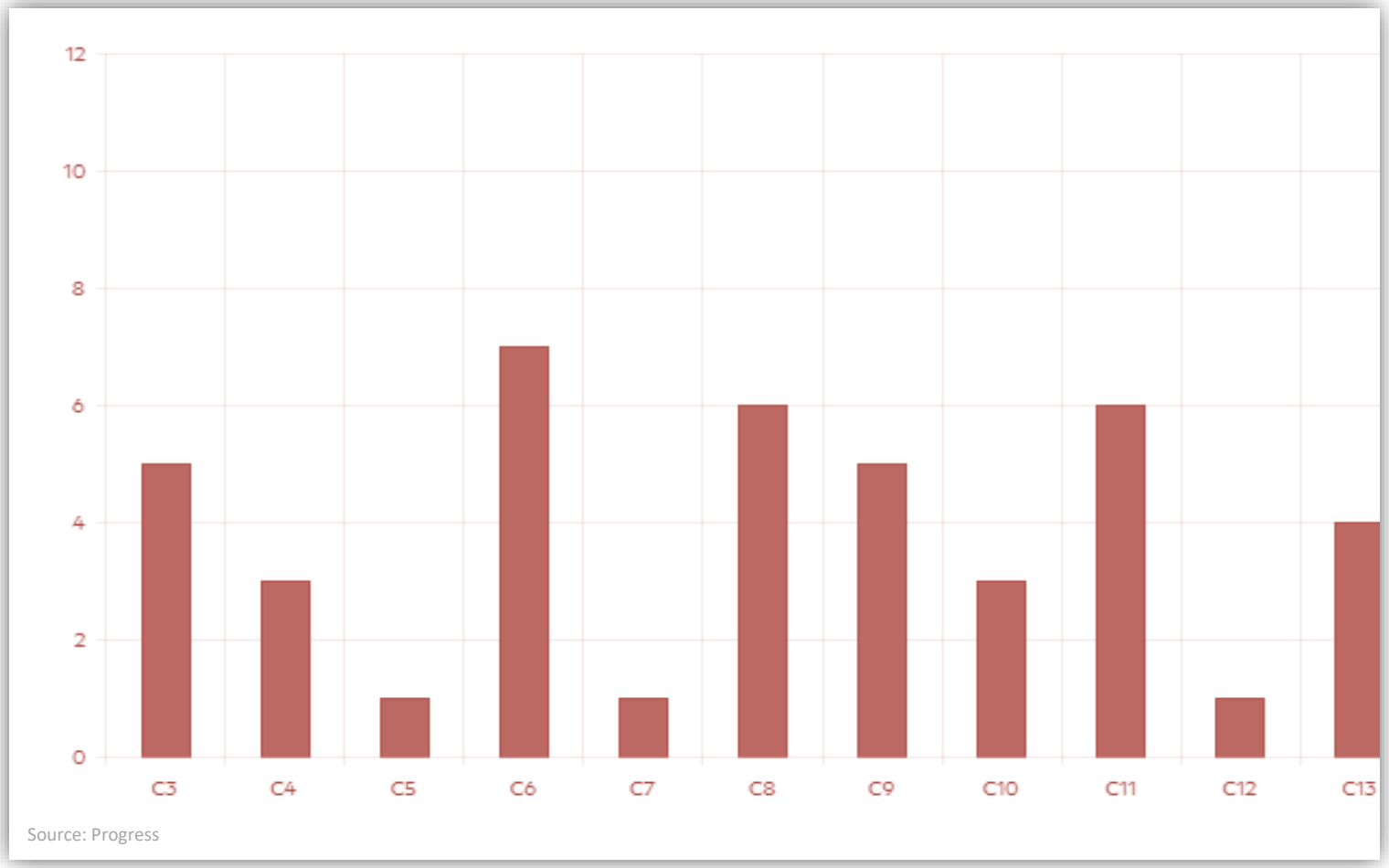
Trigger

Drag with the mouse or finger.

Action

The view window scrolls through the visualization.

Related to: Optical zoom.



Navigation: Time as a third dimension

Objective

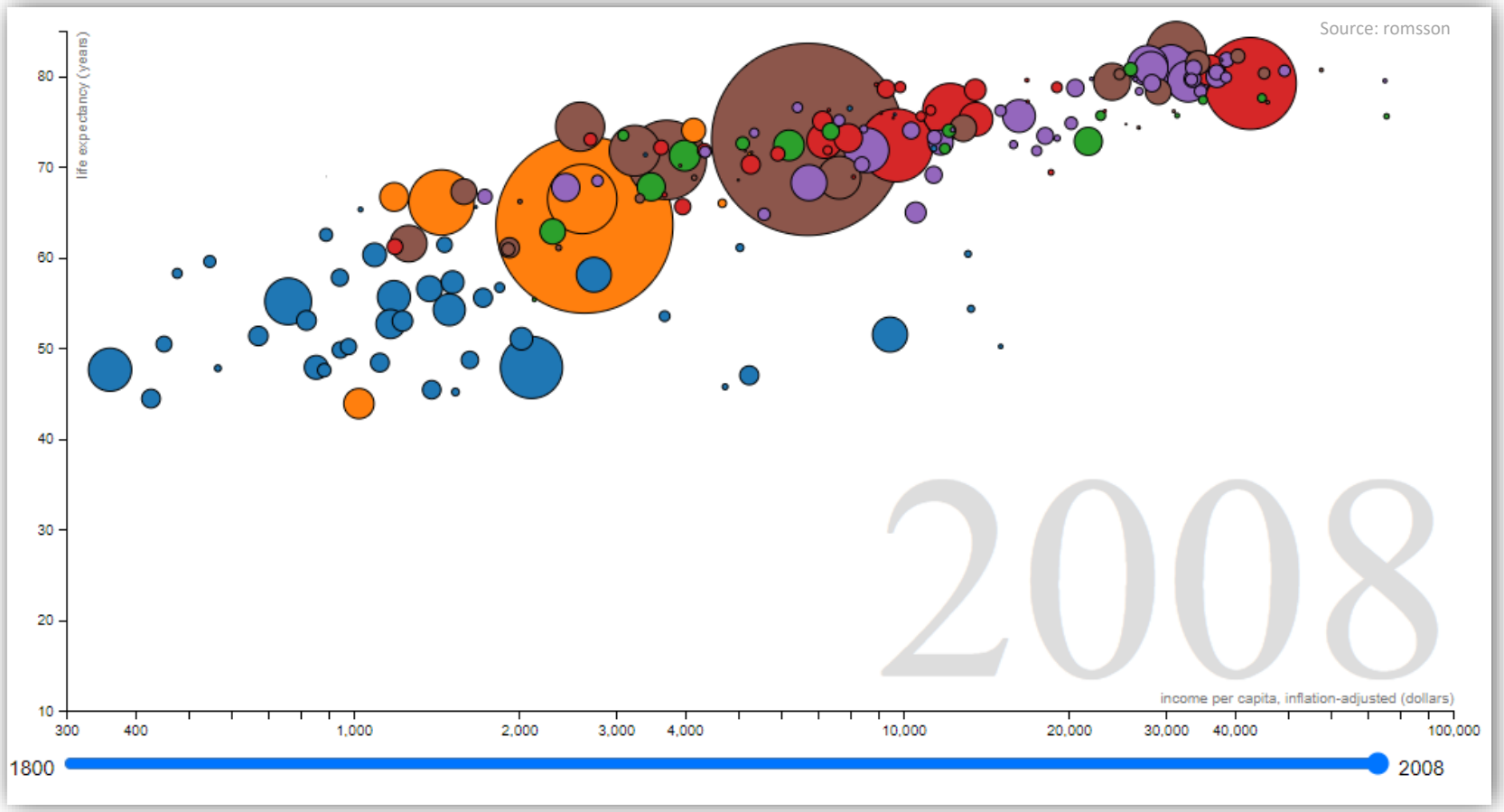
Change the display over time.

Trigger

Select box or "Play" button

Action

The display is updated with the data from the new date.



Problem solving



Problem solving



Linked views

Different charts share domain or selection.



Focus and context

Look more closely at a specific area while viewing the whole thing.



Overview and detail

There are 2 views, a general one of the whole set and a detailed one.



Details on demand

Offer 2 levels of information and show the second level only when users ask for it.

Linked views

Objective

Different charts share domain or selection.

Trigger

An action in one of the views.

Action

The action in one view (filter, drag, select...) affects the others.



Focus and context

Objective

Look more closely at a specific area while viewing the whole thing.

Trigger

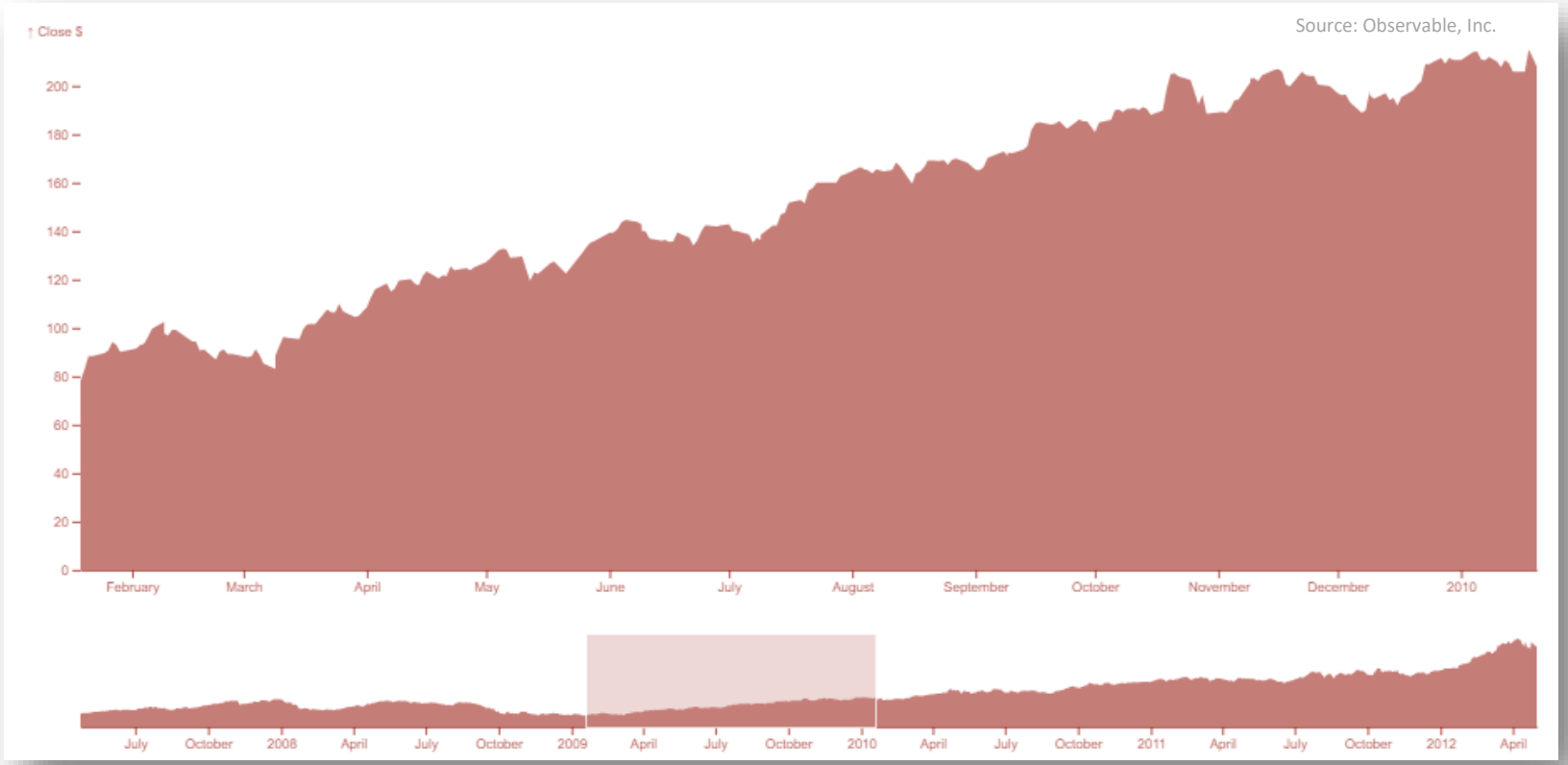
Brushing, filtering.

Action

The area that receives focus is shown in more detail.

Related to:

Zoom, linked views.



Overview and detail

Objective

Overview first, details on demand. The user can get a detailed view on demand.

Trigger

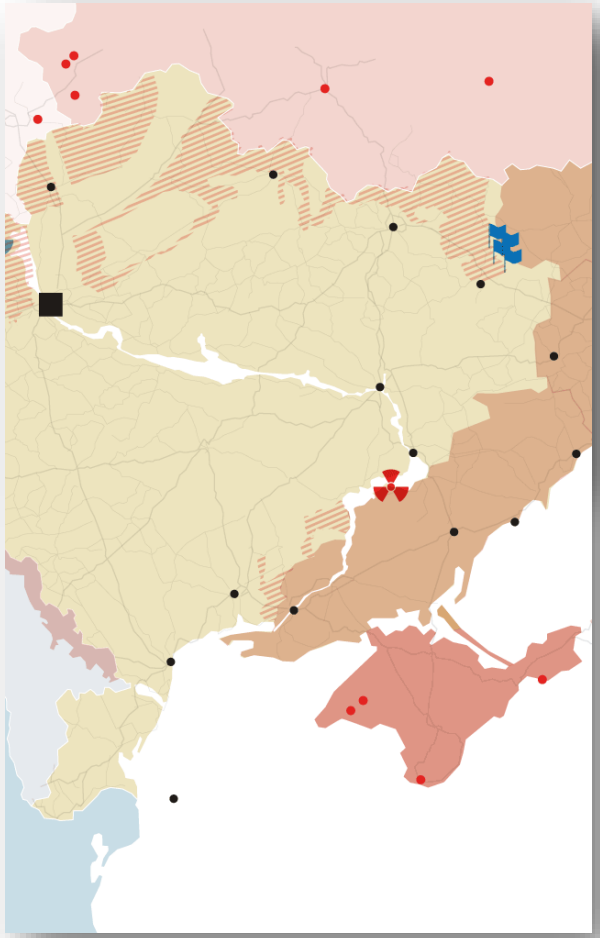
Zoom and navigation in detail.

Action

A selection triggers a zoomed version.

Related to:

Zoom, drill down.



Source: Abc.

Details on demand

Objective

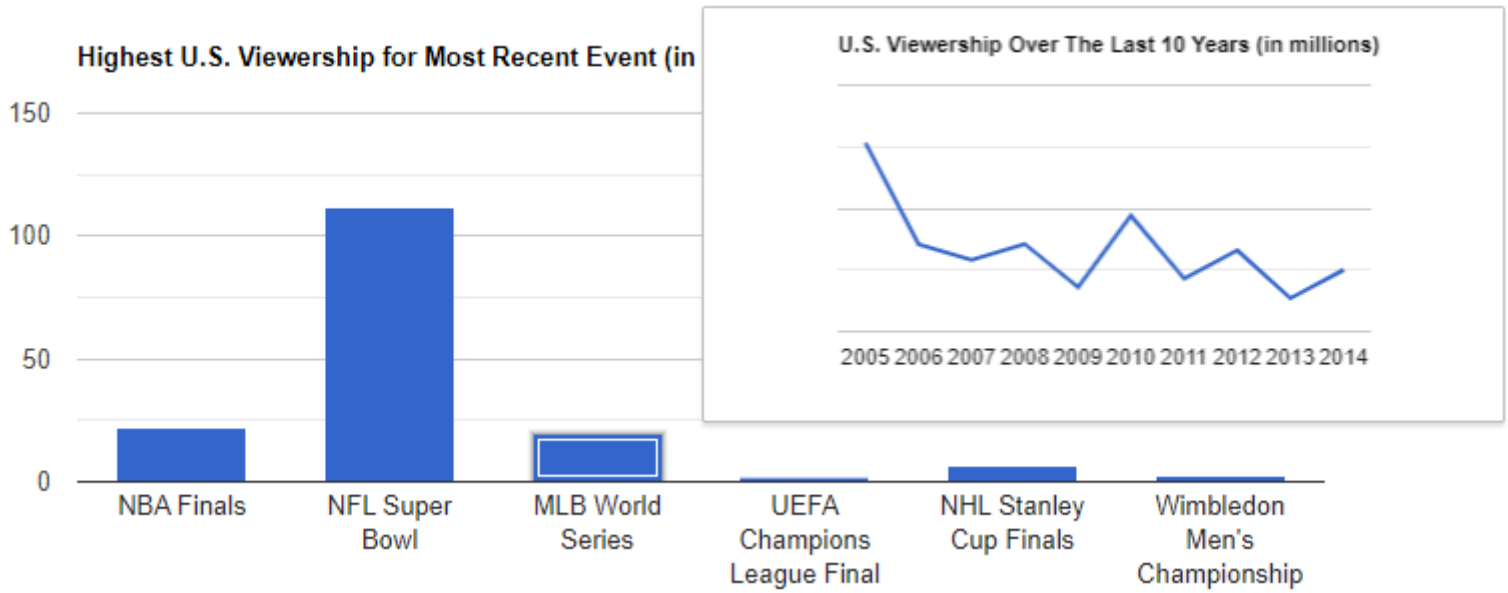
Offer 2 levels of information and show the second level only when users ask for it. It's a variation of overview and detail.

Trigger

Select, hover over.

Action

A tooltip or information box related to the selected item appears or the displayed information is updated.



Source: Google Developers

Related to:

Overview and detail, drill down.

Final Thoughts



Final Thoughts

Interaction involves an effort on the part of the user.

1. Interaction patterns may **not be known**, so:
    - Include instructions
    - Follow the conventions
    - Give immediate feedback
2. The interaction is not free: the user must decide **the action to perform and reposition himself** when the view changes.
3. Plan the interaction to be within the problem-solving natural flow.

Key ideas

1. In the digital environment, interaction offers new possibilities.

2. With the interactions of manipulation and encoding of the data, the user chooses what to see and how to display it.

3. With the interactions of exploration and navigation the user navigates through the informational space.

4. With problem-solving interactions, the user delves into the data and relates data and details.

5. Interaction has a cognitive cost for the user.

Sources

- Ware, C. (2020) Information visualization: perception for design. Burlington: Morgan Kaufmann.

Thank you for your attention

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